

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	19 July 2026
Team ID	
Project Name	Nightjar & Co. – MERN BookStore
Maximum Marks	4 Marks

Technical Architecture

Description

Nightjar & Co. follows a three-tier architecture consisting of the Presentation Layer, Application Layer, and Database Layer. Users interact with the React frontend, which communicates with the Express.js and Node.js backend through REST APIs. The backend processes business logic, authenticates users using JWT, interacts with the MongoDB database through Mongoose, and returns JSON responses to the frontend.

Architecture Diagram

Nightjar & Co. – MERN BookStore is built using a three-tier architecture comprising the Presentation Layer, Application Layer, and Database Layer. This architecture separates the user interface, business logic, and data storage, making the application modular, maintainable, and scalable.

The Presentation Layer is developed using React.js (Vite) and provides an interactive and responsive interface where users can register, log in, browse books, search and filter the catalog, view book details, and manage their shopping cart.

The Application Layer is implemented using Node.js and Express.js, which expose RESTful APIs to process user requests, handle authentication using JWT, manage business logic, and communicate with the database.

The Database Layer uses MongoDB with Mongoose to store and manage user accounts, book information, and other application data. The backend retrieves and updates this data as required and returns JSON responses to the frontend.

This architecture ensures efficient communication between components, improves code organization, and provides a strong foundation for future enhancements such as online payments, order management, and cloud deployment.



Users



React Frontend (Vite)



- Home Page
- Login/Register
- Book Catalog
- Search & Filter
- Cart



REST API (HTTP)



Express.js + Node.js API



- Authentication
- Book Management
- Cart Logic
- Order Processing



Mongoose ODM



MongoDB Database



- Users Collection
- Books Collection
- Orders Collection

Table 1: Components & Technologies

S.No.	Component	Description	Technology
1	User Interface	Web application for browsing books, login, cart, and orders	React.js (Vite), HTML5, CSS3, JavaScript
2	Application Logic – 1	Frontend routing and UI management	React Router, Context API
3	Application Logic – 2	Backend business logic and REST APIs	Node.js, Express.js
4	Application Logic – 3	Authentication and authorization	JWT, bcrypt.js
5	Database	Stores users, books, and order information	MongoDB with Mongoose
6	Cloud Database	Not implemented (can be extended to MongoDB Atlas)	MongoDB Atlas (<i>Future Enhancement</i>)
7	File Storage	Stores static assets and uploaded images	Local File System
8	External API – 1	No external APIs used in the current version	—
9	External API – 2	Not applicable	—
10	Machine Learning Model	Not implemented	—
11	Infrastructure	Local development server	Node.js Local Server

Table 2: Application Characteristics

S.No.	Characteristic	Description	Technology
1	Open-Source Frameworks	Frontend and backend developed using open-source frameworks	React.js, Node.js, Express.js, MongoDB, Mongoose
2	Security Implementations	Password hashing, JWT authentication, protected API routes	bcrypt.js, JWT
3	Scalable Architecture	Modular three-tier architecture with separation of frontend, backend, and database	MERN Stack, REST APIs
4	Availability	Accessible through modern web browsers whenever the backend server is running	Node.js Server, HTTP
5	Performance	Efficient API communication, lightweight JSON responses, optimized MongoDB queries	REST API, MongoDB, Mongoose